

SOFTWARE ENGINEERING LAB 2

Blood Bank Inventory & Emergency Donor Matcher

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A. Project Overview

The project is a Blood Bank Inventory & Emergency Donor Matcher. The system is intended to maintain real-time blood inventory information, monitor blood component shelf life, and support emergency donor matching and notifications during shortages. The project stakeholders/actors identified in the requirements material are the Emergency Requester and Blood Bank Manager.

The Jira project is organized around six epics recorded during project setup:

Epic Key	Epic
BBIM-1	Blood Inventory Management
BBIM-2	Donor Management
BBIM-3	Emergency Donor Matching
BBIM-4	Blood Request & Emergency Management
BBIM-5	Notifications & Alerts
BBIM-6	Reports & Administration

Note: The supplied Jira evidence shows BBIM-1, BBIM-3 and BBIM-4 at 100% epic progress, while the project summary reports 32 total work items, 26 completed and 6 To Do. The screenshots do not provide enough evidence to assign all stories to every epic or to state that BBIM-5 and BBIM-6 contain stories.

B. Epic and User Story Inventory

The Jira burnup/activity evidence explicitly lists the following BBIM user stories. Story-point values below are the values evidenced by the burnup event sequence for Sprint 1. A value of 0 means that the supplied event evidence did not add story points for that item at that stage; it is not interpreted as a claim that the story is inherently zero effort.

Issue	User Story	Story Points	Evidence/Scope
BBIM-7	Blood Inventory Tracking	0	Sprint 1
BBIM-8	Record blood inventory	3	Sprint 1
BBIM-9	Update blood inventory	5	Sprint 1
BBIM-10	Monitor blood expiry	3	Sprint 1
BBIM-11	Register donor	3	Sprint 1
BBIM-12	Maintain donor eligibility	5	Sprint 1
BBIM-13	Update donor availability	0	Sprint 1
BBIM-14	Find compatible emergency donors	5	Sprint 1
BBIM-15	Match donors by blood group	0	Sprint 1
BBIM-16	Prioritize emergency blood requests	0	Sprint 1
BBIM-17	Notify compatible donors	3	Sprint 1
BBIM-18	Track emergency matching status	0	Sprint 1
BBIM-19	Record donor response	3	Sprint 1
BBIM-20	Create emergency blood request	5	Sprint 1
BBIM-21	View emergency blood request details	0	Sprint 1
BBIM-22	Update emergency blood request status	3	Sprint 1
BBIM-23	Notify blood bank manager about	0	Sprint 1

	emergency request		
BBIM-24	Update blood inventory levels	0	Later added/removed from sprint
BBIM-25	View current blood inventory	0	Later added/removed from sprint
BBIM-26	Record blood component details	0	Later added/removed from sprint

Observed Sprint 1 committed scope: 38 story points. The burndown report uses 38 as the sprint work-scope line.

1. Story Points and Planning Poker Estimation

Story-point estimation was handled in Jira using the Story Points field. The supplied burndown evidence shows a Sprint 1 work scope of 38 story points. The completion sequence increases the completed total from 0 to 38 as stories are marked complete.

The evidence also shows later estimate changes: BBIM-26 changed from 38 to 41, BBIM-25 from 41 to 43, BBIM-24 from 43 to 46, and BBIM-23 from 46 to 48. These items were subsequently removed from the sprint, returning the sprint scope to 38. This is important because the final sprint burndown scope is 38, not 48.

This report does not invent individual Planning Poker votes because no Planning Poker voting screen was supplied. It records only the story-point changes visible in Jira.

2. Prioritization and Backlog Ranking

The supplied Jira screenshots confirm that the backlog and sprint were managed in Jira, but they do not show the complete priority column or the final ranked backlog order. Therefore, this report does not fabricate priority labels. The story list is presented in the order evidenced by the Sprint 1 completion sequence.

For the submitted evidence, the most useful ranking is the actual completion sequence visible in the burndown report: BBIM-7, BBIM-9, BBIM-11, BBIM-8, BBIM-10, BBIM-12, BBIM-13, BBIM-14, BBIM-15, BBIM-16, BBIM-17, BBIM-18, BBIM-19, BBIM-20, BBIM-21, BBIM-22 and BBIM-23, followed by estimate changes and removal of BBIM-24 to BBIM-26.

3. Sprint 1

Jira shows BBIM Sprint 1 with the date range September 1, 2026 – September 8, 2026. The burndown report identifies 38 story points as the sprint work scope. The Sprint 1 activity log records the sprint start and the completion of the listed BBIM stories.

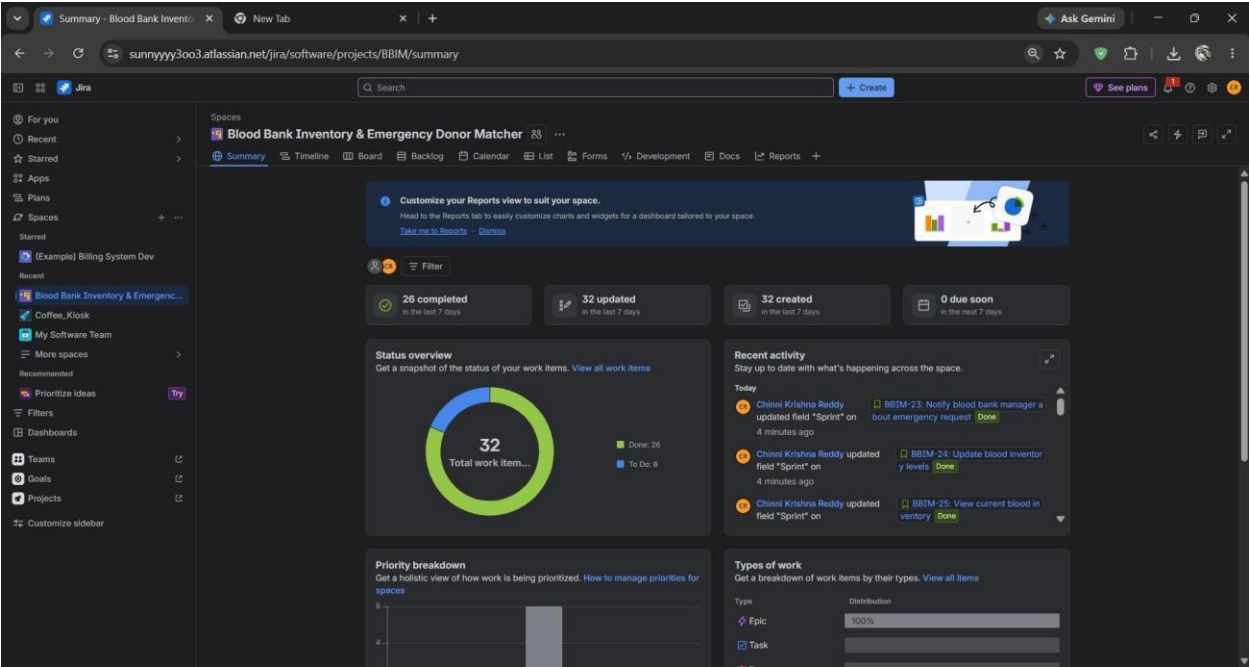
The sprint was used as a mock execution cycle. Work items were marked completed in Jira, allowing the burndown report to track the accumulated completed story points against the sprint scope.

4. Sprint Progress and Activity

The supplied Jira evidence shows the work moving to completion rapidly during Sprint 1. The activity report records individual work-item completions and the corresponding story-point totals. It also records a re-open/re-completion event for BBIM-23 and later estimate updates/removals for BBIM-23 to BBIM-26.

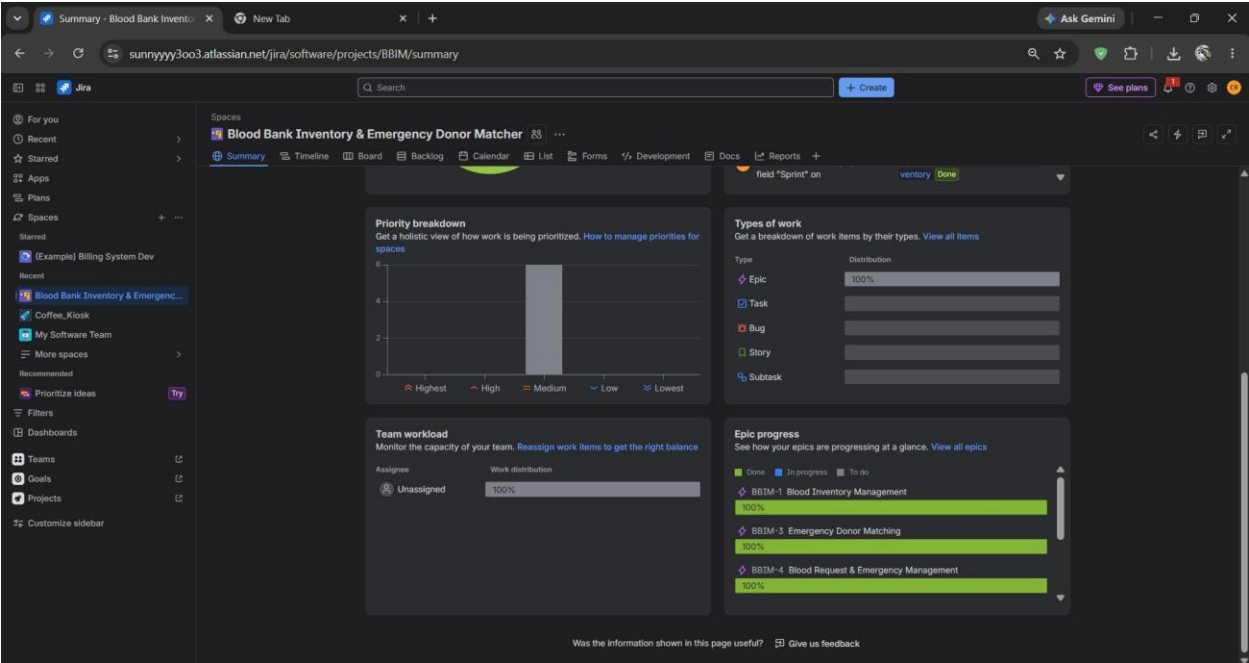
The screenshots supplied for this report do not display the text of progress comments. Consequently, no comment text has been invented. The evidence is reported exactly as visible: work-item completion, re-opening, estimate changes, and removal from sprint.

Output Screenshot 1 – Jira Project Summary



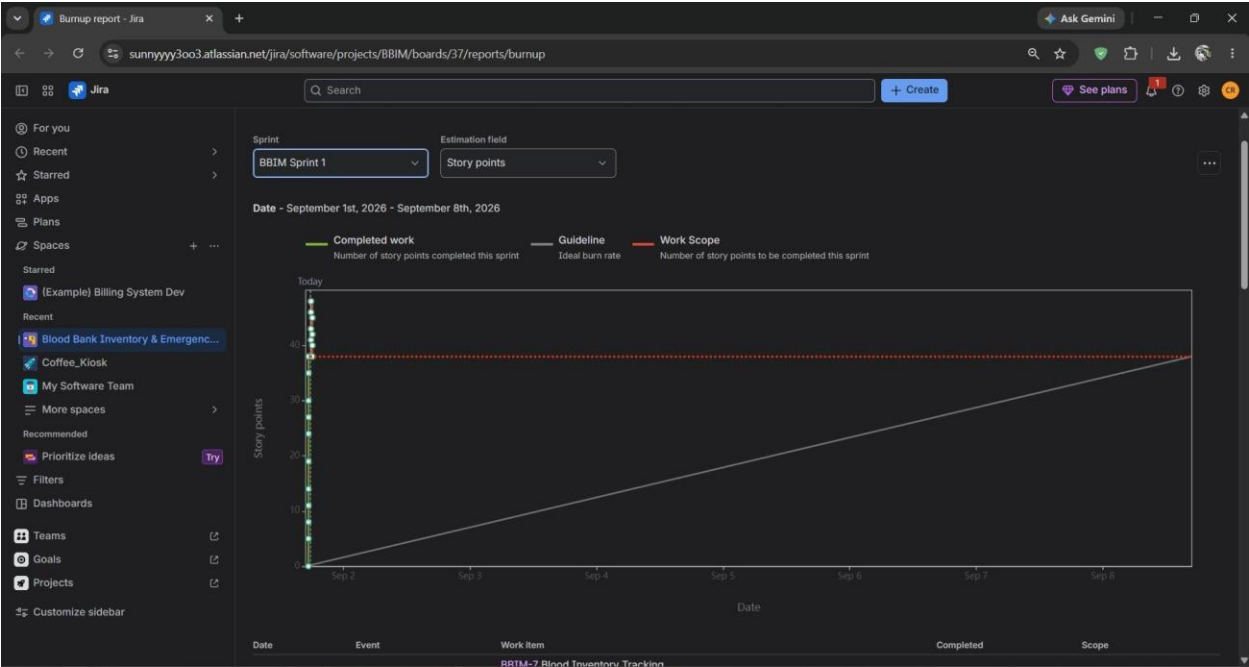
Jira project Summary showing 32 total work items, 26 completed and 6 To Do, with recent activity visible.

Output Screenshot 2 – Jira Project Summary: Lower Section



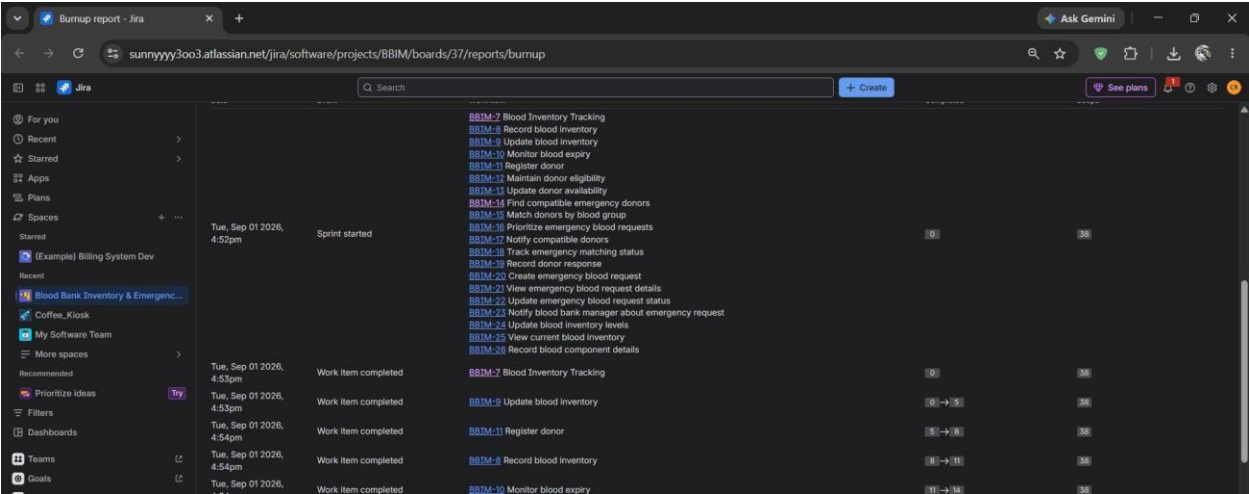
Priority breakdown, types of work, team workload and epic progress.

Output Graph 1 – Jira Sprint 1 Burndown Chart



Sprint 1 burndown chart for September 1–8, 2026, with a 38-story-point work-scope line.

Output Screenshot 3 – Jira Sprint 1 Activity Log



The screenshot displays the Jira interface for a sprint titled "Burnup report - Jira". The left sidebar shows navigation options like "For you", "Recent", "Starred", "Apps", "Plans", "Spaces", and "Dashboards". The main content area shows a list of work items with columns for date, status, and details. The first entry is "Sprint started" on "Tue, Sep 01 2026, 4:52pm". This is followed by a list of completed work items, each with a status of "Work item completed" and a time of "4:53pm" or "4:54pm". The work items are linked to various issues, including "BBIM-7 Blood Inventory Tracking", "BBIM-8 Record blood inventory", "BBIM-9 Update blood inventory", "BBIM-10 Monitor blood expiry", "BBIM-11 Register donor", "BBIM-12 Maintain donor eligibility", "BBIM-13 Update donor availability", "BBIM-14 Find compatible emergency donors", "BBIM-15 Match donors by blood group", "BBIM-16 Prioritize emergency blood requests", "BBIM-17 Notify compatible donors", "BBIM-18 Track emergency matching status", "BBIM-19 Record donor response", "BBIM-20 Create emergency blood request", "BBIM-21 View emergency blood request details", "BBIM-22 Update emergency blood request status", "BBIM-23 Notify blood bank manager about emergency request", "BBIM-24 Update blood inventory levels", "BBIM-25 View current blood inventory", and "BBIM-26 Record blood component details".

Date	Status	Details
Tue, Sep 01 2026, 4:52pm	Sprint started	BBIM-7 Blood Inventory Tracking BBIM-8 Record blood inventory BBIM-9 Update blood inventory BBIM-10 Monitor blood expiry BBIM-11 Register donor BBIM-12 Maintain donor eligibility BBIM-13 Update donor availability BBIM-14 Find compatible emergency donors BBIM-15 Match donors by blood group BBIM-16 Prioritize emergency blood requests BBIM-17 Notify compatible donors BBIM-18 Track emergency matching status BBIM-19 Record donor response BBIM-20 Create emergency blood request BBIM-21 View emergency blood request details BBIM-22 Update emergency blood request status BBIM-23 Notify blood bank manager about emergency request BBIM-24 Update blood inventory levels BBIM-25 View current blood inventory BBIM-26 Record blood component details
Tue, Sep 01 2026, 4:53pm	Work item completed	BBIM-7 Blood Inventory Tracking
Tue, Sep 01 2026, 4:53pm	Work item completed	BBIM-8 Update blood inventory
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-11 Register donor
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-8 Record blood inventory
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-10 Monitor blood expiry

Sprint start and the first sequence of completed stories are recorded.

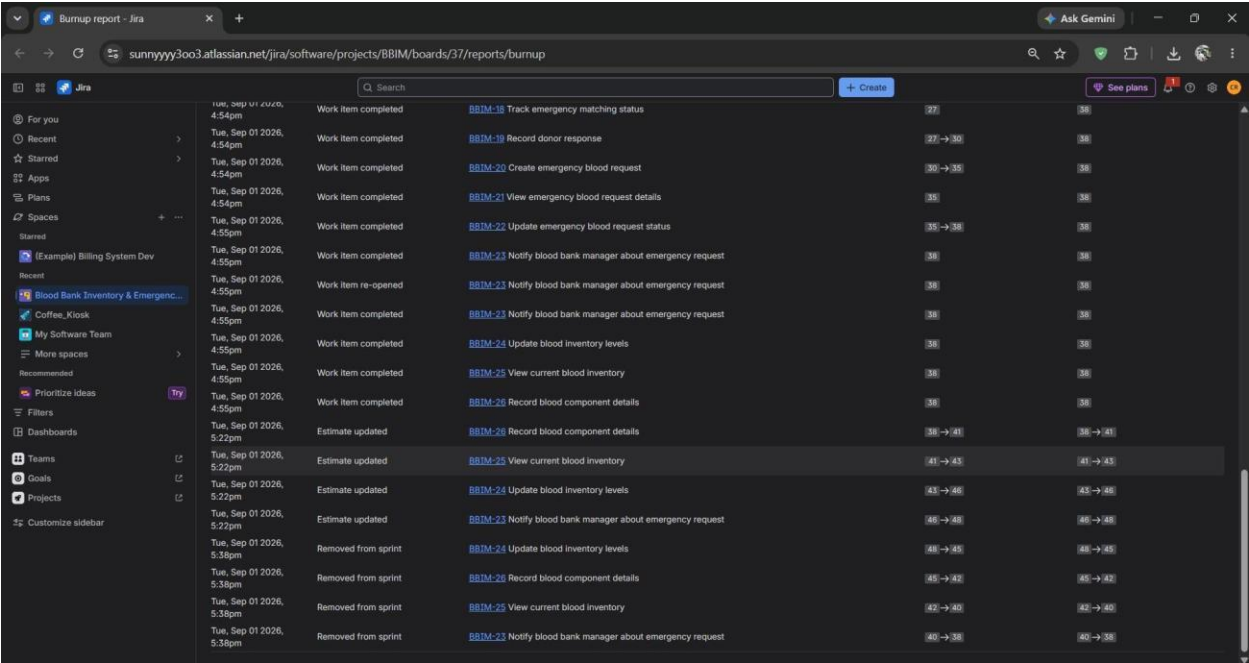
Output Screenshot 4 – Jira Sprint 1 Activity Log

The screenshot displays the Jira interface for a sprint report. The left sidebar shows navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Filters'. The main content area lists activity log entries. Each entry includes a timestamp, a status (e.g., 'Work item completed'), a link to the work item, and a progress bar. The work items are BBIM-12 through BBIM-26, with BBIM-24 highlighted. The progress bars show the completion status of each item, with BBIM-12 through BBIM-23 being fully completed and BBIM-24 through BBIM-26 showing partial completion.

4:54pm	Work item completed	BBIM-12 Maintain donor eligibility	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-13 Update donor availability	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-14 Find compatible emergency donors	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-15 Match donors by blood group	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-16 Prioritize emergency blood requests	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-17 Notify compatible donors	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-18 Track emergency matching status	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-19 Record donor response	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-20 Create emergency blood request	
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-21 View emergency blood request details	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-22 Update emergency blood request status	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-23 Notify blood bank manager about emergency request	
Tue, Sep 01 2026, 4:55pm	Work item re-opened	BBIM-23 Notify blood bank manager about emergency request	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-23 Notify blood bank manager about emergency request	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-24 Update blood inventory levels	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-25 View current blood inventory	
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-26 Record blood component details	
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-26 Record blood component details	
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-25 View current blood inventory	
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-24 Update blood inventory levels	

Completion sequence continues through BBIM-14 to BBIM-26.

Output Screenshot 5 – Jira Sprint 1 Activity Log



The screenshot displays the Jira interface for a project named 'Burnup report - Jira'. The left sidebar shows navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Filters'. The main content area shows a list of activity log entries for the sprint. Each entry includes a timestamp, a description of the event, and a link to the related issue.

Timestamp	Event	Issue Link	From	To
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-18 Track emergency matching status	32	38
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-19 Record donor response	37 → 38	38
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-20 Create emergency blood request	35 → 35	38
Tue, Sep 01 2026, 4:54pm	Work item completed	BBIM-21 View emergency blood request details	35	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-22 Update emergency blood request status	35 → 38	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-23 Notify blood bank manager about emergency request	38	38
Tue, Sep 01 2026, 4:55pm	Work item re-opened	BBIM-23 Notify blood bank manager about emergency request	38	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-23 Notify blood bank manager about emergency request	38	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-24 Update blood inventory levels	38	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-25 View current blood inventory	38	38
Tue, Sep 01 2026, 4:55pm	Work item completed	BBIM-26 Record blood component details	38	38
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-26 Record blood component details	38 → 41	38 → 41
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-25 View current blood inventory	41 → 43	41 → 43
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-24 Update blood inventory levels	43 → 46	43 → 46
Tue, Sep 01 2026, 5:22pm	Estimate updated	BBIM-23 Notify blood bank manager about emergency request	46 → 46	46 → 46
Tue, Sep 01 2026, 5:38pm	Removed from sprint	BBIM-24 Update blood inventory levels	46 → 46	46 → 46
Tue, Sep 01 2026, 5:38pm	Removed from sprint	BBIM-26 Record blood component details	46 → 46	46 → 46
Tue, Sep 01 2026, 5:38pm	Removed from sprint	BBIM-25 View current blood inventory	46 → 46	46 → 46
Tue, Sep 01 2026, 5:38pm	Removed from sprint	BBIM-23 Notify blood bank manager about emergency request	46 → 38	46 → 38

Later estimate updates and removal-from-sprint events are visible.

5. Sprint Summary

The Jira evidence shows Sprint 1 completing the committed 38 story points. The burndown activity reaches 38/38 before the later estimate changes and removals. This demonstrates that the original sprint scope was fully completed in the evidence provided.

The project Summary page also reports 26 completed work items and 6 To Do at the time of capture. Because that project-level count includes work outside the original 38-point Sprint 1 scope, it should not be confused with the 38 story points completed in the sprint.

A notable activity was BBIM-23, which was re-opened and then completed again. The report therefore reflects the actual Jira history rather than treating the activity log as a perfectly linear sequence.

6. Burndown Chart

The Sprint 1 burndown chart plots completed story points against the sprint period and displays a 38-point work-scope line. The visible completed-work markers rise from 0 to 38. The ideal guideline runs from the starting scope toward zero over the sprint period.

Because the work was completed very quickly in the captured Jira history, the completed-work markers are concentrated near the beginning of the displayed date range. The chart therefore provides direct evidence that the committed 38 story points were completed.

7. Group Reflection

During this lab, we learned how to create and organize an Agile backlog in Jira using Epics and User Stories. We prioritized the user stories based on their importance and assigned Story Points using the Fibonacci sequence. We also learned how Planning Poker can be used for estimation.

We created and simulated a Sprint by moving user stories through To Do → In Progress → Done and added comments to track progress. The Burndown Chart helped us understand the amount of work completed and the work remaining during the sprint. Overall, this lab helped us understand how Agile Scrum practices can be applied using Jira for managing software development work.

The Jira exercise demonstrates the practical use of an Agile backlog and sprint workflow for the Blood Bank Inventory & Emergency Donor Matcher project. Creating separate epics and stories helped divide the system into manageable functional areas. Assigning story points made the sprint scope measurable, while the burndown report provided a visual view of progress.

The activity history also shows why estimates and sprint scope need to be controlled carefully. BBIM-23 to BBIM-26 were involved in later estimate changes, and BBIM-24 to BBIM-26 were removed from the sprint. Keeping the final sprint scope at 38 points made the burndown interpretation consistent.

Overall, the supplied evidence demonstrates the core Lab 3 workflow: Jira work-item creation, sprint execution, progress tracking, story-point accounting, scope changes, and burndown reporting.

Appendix – Requirements Basis

The project requirements material describes the system as a real-time blood-bank stock management and emergency donor-matching system. Key requirements include cross-referencing emergency requests against inventory, managing inventory details and expiry dates, identifying eligible donors using compatibility/location/eligibility criteria, maintaining donor profiles, and generating reports.

Non-functional requirements include transactional consistency and 24/7 availability under emergency load.

These requirements provide the domain basis for the Jira epics and stories, while the Jira screenshots provide the execution evidence used in this report.